

Madan Bhandari College of Engineering
Urlabari-3, Morang

Final Internal Examination (Fall 2025)

Level: Bachelor

Full Marks: 100

Programme: B.E (Computer)

Pass Marks: 45

Year/Part:- IV/I

Time: 3 hrs

Subject: - Engineering Economics (Set A)

- ✓ Candidates are required to give their answers in their own words as far as possible
- ✓ Attempt all questions

1. a). why an engineer should follow fundamental principles of engineering economy? Explain. 7.

b). You make 10 annual deposits of 1000 each into a bank account paying 5% interest per year. The first deposit will be made one year from today. How much money can be withdrawn from this account after 10 years? What is the present worth of that sum? 8

2. a). Write about the types of cost involved in cost accounting? Clarify the concept of cost for business decision making. 7

b). Determine conventional and modified B/C ratio for the given project. The cash flows are as follows:

Initial cost	Rs 4,00,000
Life of the project	10 years
Salvage benefits	100,000
Annual benefits	75000
Annual disbursements (O & M)	25000
Interest rate	10% per year

8

whether the projects is feasible or not?

3. a). Define payback period method. Explain simple and discounted payback period method with a suitable example.

b). A mobile company is taking quotations for purchase, installation and operation of microwave towers for long period. If MARR is 15% determine the best alternative project using repeatability assumption and PW method. 8

Quotations	Project A	Project B
Equipment cost (Rs)	6500000	5800000
Installation (Rs)	1500000	20,00,000
Annual maintenance (Rs)	100000	125000

Annual extra charge (Rs)	0	50000
Salvage value (Rs)	0	0
Life (yrs)	15	10

4. a). Select the best project by using BCR method using FW formulation when MARR is 8.5%. Use increment analysis if necessary. 8

Year	Project A	Project B
Initial Investment	6,00,000	10,00,000
Annual Revenue	3,00,000	3,50,000
Life	6	6
Salvage Value	1,20,000	1,80,000

- b). what is break even Analysis? Phewa cement shop open the market to sell 10,000 bags of cement in one month, the shop purchase the cement at the rate of 650 in bulk and sell at the rate of Rs 800. What will be the break-even quantity, if the shop sold all the quantities what will be the profit or loss, when the shop pay Rs 15,000 for rent and Rs 5000 for salary? Initial investment was Rs 65,00,000. 7

- 5 a). List out causes of depreciation. A machine costing of Rs 10,000 is estimated salvage value of Rs 500 at the end of 5th year. Find depreciation each year by using

- Sinking fund
- Declining balance
- SOYD method

8

- b). Discuss economic theory and ecological footprint. What role an engineer should play for overcoming ecological limit and achieve global sustainable goals? 7

- 6 a). What is financial ration? Explain major financial ratios that can be applied to decision making process in business. 7

- b). What are the funding mechanism of Nepal government to the project while tal

king about project funding mechanism?

Are you agree existing mechanism or needs to improve it? 8

7. Write short notes on: (Any two) 10

- Balance sheet or VAT
- Ecological limit and sustainable development
- Project risks

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Subject: - Engineering Economics (Set B)

✓ Candidates are required to give their answers in their own words as far as possible.

Attempt all the Questions.

1. ✓ a). Describe engineering economics. Explain the different principles of engineering economics. 5+2.
b). If you deposited Rs 50,000 the beginning of each for 7 years. How much money will be in your account at the end of 10th year when rate of interest is 6% compounded quarterly. Make also cash flow diagram. 8
2. a). Discuss any four basic cost concepts used in engineering economics. Explain why these cost concepts are important in engineering economic decisions. 7
b). A piece of new equipment has been proposed by engineers to increase the productivity of a certain manual welding operation. The investment cost of equipment is Rs 1,00,000 and has market value of Rs. 20,000 at the end of five years. The equipment will generate Rs 32,000 per year after operation costs. The company is willing to accept the project if it will earn at least 20% per year. Evaluate IRR of this equipment and suggest whether the equipment is feasible or not? 8
3. a). Find both types of BC ratios from the following project and also suggest if it is feasible or not. 7
Initial investment Rs 1500000
Annual revenue Rs 300000
Annual cost Rs 60000
Estimated useful life 8 years,
Salvage value Rs 120000
MARR 12%
b). The following data have been estimated for two feasible investment machine X and Machine Y having different useful life. If minimum attractive rate of return is 10% choose the best one project using PW method. Use repeatability assumption.

	Machine X	Machine Y
Initial Cost	20,000	30,000
Uniform annual benefit	10,000	10,000
Salvage value	2000	10000
Useful life	3	6

4. a). Select the best project by using BCR method using FW formulation when MARR is 8.5%.
Use increment analysis if necessary. 8

Year	Project A	Project B
Initial Investment	6,00,000	10,00,000
Annual Revenue	3,00,000	3,50,000
Life	6	6
Salvage Value	1,20,000	1,80,000

- b). Describe Inflation. What do you understand by Actual Dollar and Constant dollar method explain with your own example. 7
- 5 a). ABC Company is planning to purchase a machine costing of Rs 50,00,000 having estimated salvage value of Rs 45,000 and useful life 6 years. Calculate every year depreciation amount and book value under sinking fund and SOYD depreciation method, if MARR is 12%. 7
- b). Discuss economic theory and ecological footprint. What role an engineer should play for overcoming ecological limit and achieve global sustainable goals? 8
- 6 a). What is financial ration? Explain major financial ratios that can be applied to decision making process in business. 7
- b). What are the funding mechanism of Nepal government to the project while talking about project funding mechanism? 8
- Are you agree existing mechanism or needs to improve it?
7. Write short notes on: (Any two) 10
- Tax
 - Inflation and depreciation
 - Project risks